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Hone: Mining Agent Trajectories to Optimize AGENTS.md via Cross-Examination

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Code

Can a coding agent learn from sessions that already happened?

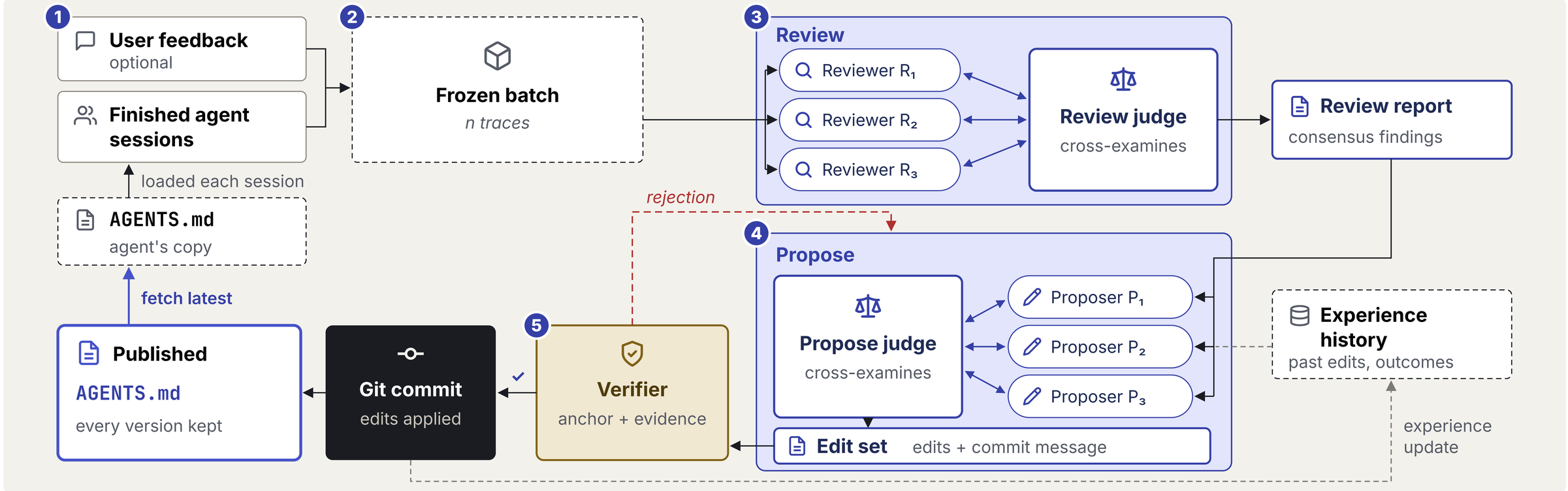
`AGENTS.md` is the instruction file a coding agent reads at session start.
Real sessions happen once, with at most a free-text comment.

Skill optimizers learn from runs labeled with a score or pass/fail, and most re-execute tasks to check each edit. Hone needs neither. It edits `AGENTS.md` from finished sessions, with or without free-text feedback.

	SkillOpt	SkillOpt-Lite	Trace2Skill	EvoSkill	Hone
Re-executes tasks	yes	yes	no	yes	**no**
Feedback signal	score	score	pass/fail	score	**any text**

① How Hone works

MAJICS, Multi-Agent Judgment through Interactive Cross-examination and Synthesis



1 Trace collection. Sessions arrive as traces with optional user feedback.

2 Batch assembly. Unanalyzed traces form a batch that stays frozen for the whole run.

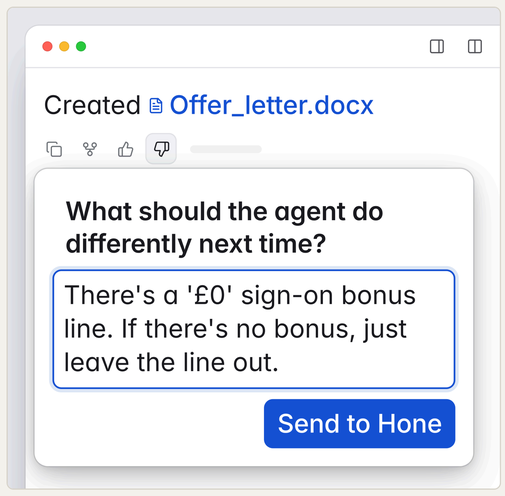
3 Review. Reviewers on different LLMs analyze the batch, and a judge cross-examines them.

4 Propose. Proposers read the review and the experience history. A second judge merges their edits.

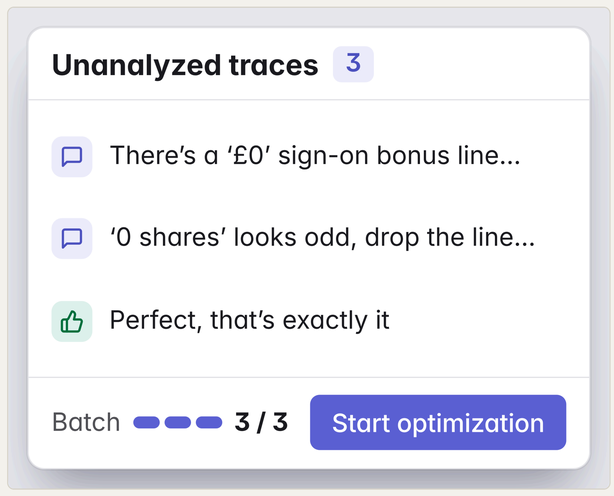
5 Verify and commit. Each edit needs a unique anchor and batch evidence, or only Propose reruns.

A finding survives only if it repeats across sessions and holds up when the judge questions it.

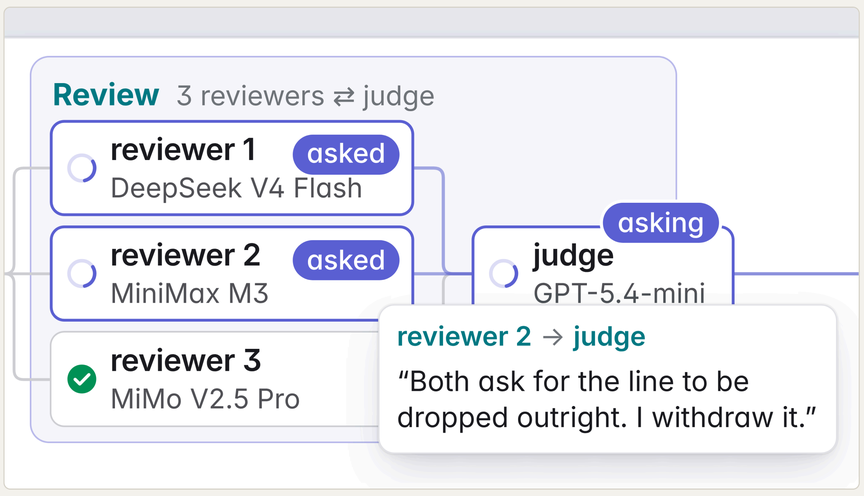
② What the demo shows



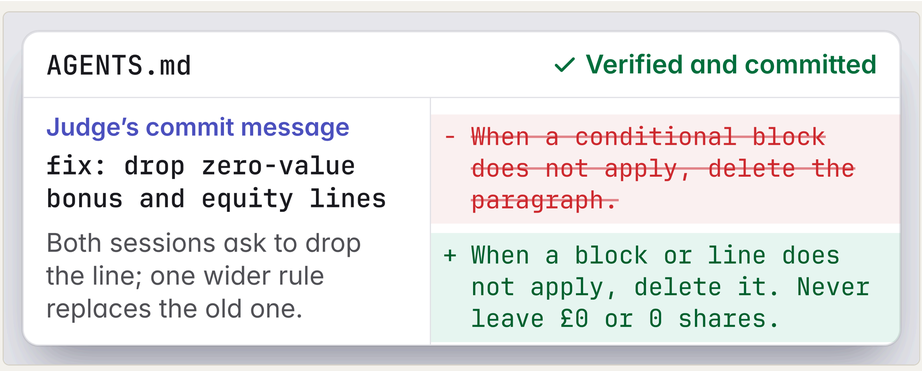
A You add feedback to the session.



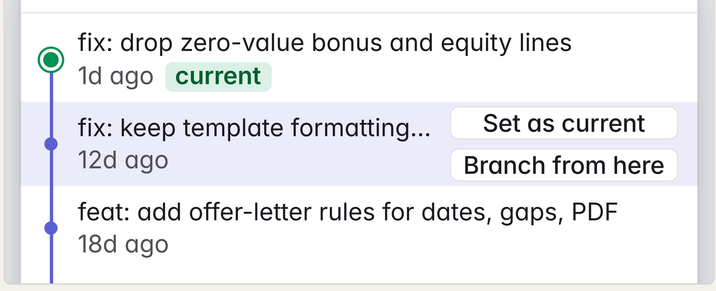
B Traces fill a batch and you start a run.



C Under questioning, one reviewer withdraws its claim.



D Hone commits the verified edit.

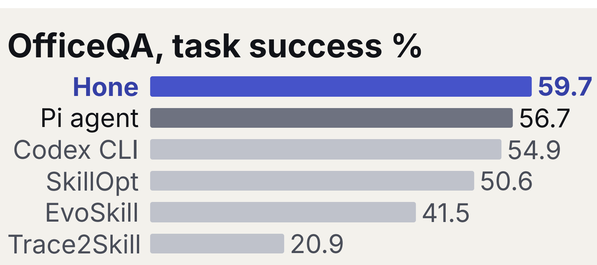


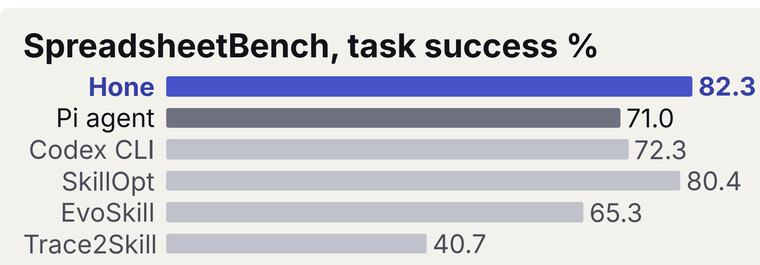
E Every version stays in Git.

Cross-examination (C). When reviewers name different causes, the judge asks each to answer the other. The goal is a better-grounded consensus than a single-round vote.

Review, then propose. Reviewers diagnose and proposers edit, so no fix comes before the diagnosis.

③ Results on two benchmarks





Hone scored highest on both benchmarks, **3.0 and 11.3 points** above the unoptimized Pi agent. Hone edited only the Pi agent's `AGENTS.md`.

All systems used GPT-5.4-mini as the target model. Hone saw only traces and expected answers. It optimizes one `AGENTS.md` per domain.

Code github.com/AaLexUser/hone

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Compared systems. Pi (Zechner, 2025), Codex CLI (OpenAI, 2025), SkillOpt (Yang et al., 2026), SkillOpt-Lite (Shen et al., 2026), Trace2Skill (Ni et al., 2026), EvoSkill (Alzubi et al., 2026). Benchmarks. OfficeQA (Opsahl-Ong et al., 2026), SpreadsheetBench (Ma et al., 2024).

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